

EEE F435 Digital Image Processing

Assignment-III

Submission Requirements

A single PDF file containing

- Student and course details
- MATLAB code (.m file) implementation.
- Screenshots of **original and processed images** for each filter.
- A short report including:
 - Explanation of each compression technique.
 - Compare different compression and decompression methods.
 - Conclusion: When would you prefer to use different compression techniques? Please provide an example for each compression.

Submission Deadline: 5:00 pm, Wednesday, 26 November 2025

Input String: "INKYPINKYPONKY"

- 1) Compress the string using the LZ-77 compression algorithm. **Window Size: 10 (search buffer size 6, lookahead buffer size 4)**
- 2) Compress the string using the LZ-78 Compression algorithm.
- 3) Compress the string using the LZW compression algorithm.
- 4) Find the compression ratio achieved by LZ-77, LZ-78, and LZW algorithms? Which algorithm has performed best on the given string? Comment.
- 5) Is there any effect of window size on the compression ratio? Comment?
- 6) Decompress the string compressed by the LZ-77 algorithm.
- 7) Write the MATLAB Code to perform LZ-77, LZ-78, and LZW compression.

Information for Q2:

Image 1--- Assignment 1--- binarize (Image 1) ----- Input Image 1

Image 1--- Assignment 1---Gray scale (Image 1) ----- Input Image 2

Image 2--- Assignment 2--- binarize (Image 2)----- Input Image 3

Image 2--- Assignment 2---Gray scale (Image 2) ----- Input Image 4

Use a Patch size of 16 x 16. Use Q=5 for gray-scale images.

Q2. Provide a MATLAB code along with a compression ratio achieved on above all four images using

- 1) Block Transform Coding
- 2) Predictive Coding
- 3) Wavelet Coding

Find the compression ratio achieved by Block Transform coding, Predictive Coding, and Wavelet Coding.

Decompress the image compressed by Block Transform coding, Predictive Coding, and Wavelet Coding and calculate the loss during compression and decompression.

Which coding performed best on the given image?

Note: Comment by a) comparing all binary images, b) comparing all grayscale images, and c) comparing binary vs. grayscale images.